

**SINGLE SOURCE or BRAND NAME DETERMINATION USING
SIMPLIFIED ACQUISITION PROCEDURES (SAP)
FOR AN ACTION NOT EXCEEDING THE SIMPLIFIED ACQUISITION THRESHOLD**

Coded Acoustic Receiver with Acoustic Release SEFSC

- 1. Agency and contracting activity.** Department of Commerce, NOAA Acquisition and Grants Office (AGO), NMFS Office of Management and Budget for the Southeast Fisheries Science Center, Miami, FL 33149
- 2. Description of supplies or services required to meet agency needs (including the estimated value).**

Estimated Total Cost of the firm fixed price purchase order: Total [REDACTED] including shipping and handling

The NOAA Southeast Fisheries Science Center uses moored archival recorders to listen for fish, invertebrate, and cetacean vocalizations at sea, as well as sounds from physical and anthropogenic sources. These recorders, when deployed in waters that are too deep to dive, require acoustic release systems to be attached in order to be able to get the recorders back to the surface. The market for acoustic releases is very limited but is slowly developing. The VR2AR-69kHz Coded Acoustic Receiver with Acoustic Release and VR2AR-X-69kHz Coded Acoustic Receiver with Acoustic Release (Extended Life) are tried and tested products that deliver high performance, reliability and longevity in the least expensive, and most robust option currently on the market for our needs. Further, it is the only Acoustic Release capable of receiving, decoding and logging transmissions from active acoustic tags deployed on fish species, providing important ecosystem and trophic interaction data. NOAA scientists and collaborators have used this release with great success in remote and harsh field conditions, providing validation of the release's performance. SEFSC currently owns 63 VEMCO VR2AR-69kHz Coded Acoustic Receiver with Acoustic Releases to allow deployment and recovery of acoustic recorder moorings, which collect ocean noise and marine life acoustic data, in deep waters for this project and requires 21 additional units that are fully compatible with existing acoustic releases and transmitting deck boxes in SEFSC's inventory. The acoustic monitoring projects for endangered Rice's whales aim to deploy arrays of acoustic moorings in their occupied shelf-break habitat throughout the Gulf of Mexico 6 months to one-year at a time to evaluate long-term trends in density and distribution, and evaluate habitat use patterns.

This purchase request for 5 VEMCO VR2AR-69kHz Coded Acoustic Receiver with Acoustic Release (unit price [REDACTED]) and 16 VEMCO VR2AR-X-69kHz Coded Acoustic Receiver with Acoustic Release (Extended Life) (unit price [REDACTED]) will allow SEFSC to deploy and recover moorings with acoustic recordings that will collect acoustic data for deployments of up to 6 months and 1 year, respectively, with concurrent sampling of any fish tagged with VEMCO acoustic transmitting tags. Alongside the acoustic releases, there is a requirement for 1 VR100 Transponding Hydrophone with 50m Cable to communicate with and release the acoustic releases at the end of the deployment, and 10 VR2AR redeployment part kits (unit [REDACTED]) to allow redeployment. The acoustic release units must be able to communicate with a Vemco VR100 - 8 Channel DSP Manual Tracking Acoustic Receiver transmitting deck box with corresponding VHTx 69kHz Transponding Hydrophone to arm, and activate the acoustic release to release the moored instruments from their anchor for recovery.

3. Identification of the single source or the brand name to be solicited.

Vendor Name: InnovaSea
 Contact: Tristan Norman; tristan.norman@innovasea.com
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 Address: 20 Angus Morton Drive
 Bedford, Nova Scotia
 Canada B4B 0L9

4. Supporting rationale. Only one source or brand name is reasonably available as detailed below:

Other - Only one source is capable:

While there is competition on the market for acoustic release systems, no other brand meets our specific needs, such as; the tested and tried reliability together with the SoundTrap 600 recorder, the ease and small size of the system, and the ability to receive, decode and log acoustic transmissions from VEMCO tags. Acoustic release systems are notorious for their failure to 'release', which when attached to an acoustic recorder full of essential data is of major importance. Our NOAA colleagues have tried a number of the acoustic releases identified below, however, upon pooling our experiences, the VEMCO VR2AR remains the most reliable and cost effective. Further, it is the only acoustic release jointly capable of receiving valuable transmissions from VEMCO acoustic tags.

This is the most cost effective and most reliable of acoustic releases currently on the market, and is only manufactured and sold by this company. Releases like this make up a tiny market and typically only service specific goals of the research community and industry. Our options are limited. Further, this is the only acoustic release that does double duty by also recording the tag transmissions from acoustically tagged fish. Understanding the ecosystem makeup of Rice's whales is a key component of our research.

Innovasea are the regional VEMCO distributor, which have been leading the design and manufacturing in compact, reliable, lightweight, robust, and cost-effective releases (see IGE document). VEMCO Ltd is used extensively and has also been the technology company of choice for a variety of other leading marine research organizations and institutes globally, including; WHOI, Scripps Institution of Oceanography, University of St Andrews, University of Aarhus, and more.

Numerous university and government researchers in the Gulf of Mexico use VEMCO acoustic tags to understand the distribution and movement patterns of a variety of fish and sea turtle species in the Gulf of Mexico, with hundreds of individuals tagged. The VEMCO VR2AR acoustic releases and receivers are the only acoustic release that is compatible with these tag transmitters which have coded signals that require decoding to be used. Use of VEMCO VR2AR acoustic releases and receivers will facilitate cross-project comparisons to provide additional ecosystem context to the data being collected by our co-deployed passive acoustic recorders.

5. Market Research.

There are no alternative vendors that sell combined acoustic releases and receivers capable of both releasing moored instruments from their anchor and receiving, decoding, and logging VEMCO acoustic tag transmissions, or that have the ability to communicate with SEFSC's existing Vemco VR100 - 8 Channel DSP Manual Tracking Acoustic Receiver transmitting deck box with corresponding VHTx 69kHz Transponding Hydrophone to arm, and activate the acoustic release to release the moored instruments from their anchor for recovery. There are other potential sources of oceanographic acoustic releases, a sampling of which are included in the table below. While some of these are less expensive than the VR2AR units, none of them have all the required capabilities, including compatibility with existing SEFSC transmitting deck boxes, ability to receive VEMCO acoustic tag transmissions, ability to deploy in depths up to 500 m, release loads of at least 250 lb, a battery life greater than 1 year, and a robust and reliable mechanical release mechanism. Alternative acoustic releases are also available from other vendors (Teledyne-Benthos, Seatronics, Edgetech), but these acoustic releases are also more expensive and are not capable of concurrently receiving, decoding and logging the VEMCO acoustic tag transmissions as required for our project. Further, a lack of compatibility with the existing transmitting deck box would add additional expense as it would require the purchase of a new transmitting deck box.

Vendor Name	Contact	Item	Max Depth	Max Load	Mechanism	Unit Price
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